

UNIT IV: Packages, Interfaces & I/O

1. What is a package in Java?

A: A package is a way to group related classes and interfaces together. It's used for organization and to prevent naming conflicts.

2. What is the keyword to create a package?

A: The `package` keyword, followed by the package name. It must be the first line of the code.

3. What is the difference between `import java.util.Scanner;` and `import java.util.*;` ?

A: `import java.util.Scanner;` imports only the `Scanner` class. `import java.util.*;` imports all public classes and interfaces from the `java.util` package.

4. Explain the four access modifiers in Java.

A: `public` : Accessible from any class, anywhere.

- `protected` : Accessible within its own package and by subclasses in other packages.
- `default (no keyword)`: Accessible only within its own package.
- `private` : Accessible only within its own class.

5. What is an interface?

A: An interface is a blueprint for a class. It defines a set of abstract methods that a class must implement. It's used to achieve 100% abstraction and multiple inheritance.

6. What is the main difference between a class and an interface?

A: A class *describes* both attributes and behaviors (with implementation), while an interface only *defines* the behaviors (a contract) that a class must implement.

7. Can a class extend multiple classes?

A: No, Java does not support multiple inheritance of classes.

8. Can a class implement multiple interfaces?

A: Yes, a class can implement any number of interfaces using the `implements` keyword.

9. Can an interface extend another interface?

A: Yes, an interface can extend another interface (or multiple interfaces) using the `extends` keyword.

10. What are variables in an interface, by default?

A: All variables in an interface are, by default, `public`, `static`, and `final`.

11. What are methods in an interface, by default?

A: All methods in an interface are, by default, `public` and `abstract`.

12. Can an interface have methods with a body?

A: Yes, since Java 8, an interface can have methods with a body using the `default` or `static` keywords.

13. What is a nested interface?

A: A nested interface is an interface that is declared inside another class or another interface.

14. What is a stream in Java?

A: A stream is a sequence of data that flows from a source (like a file or keyboard) to a destination (like the console or another file).

15. What is the `File` class used for?

A: The `File` class is used to get information about a file or directory (like its path, name, or if it exists). It is *not* used for reading or writing data to the file.

16. What is the main difference between a Byte Stream and a Character Stream?

A: Byte Streams handle raw binary data (8-bit bytes), which is good for files like images or audio. Character Streams handle text data (16-bit characters) and can manage character encodings.

17. What are the top-level abstract classes for Byte Streams?

A: `InputStream` and `OutputStream`.

18. What are the top-level abstract classes for Character Streams?

A: `Reader` and `Writer`.

19. **Which stream would you use to read a text file?**

A: A Character Stream, like `FileReader`, because it correctly handles character-based data.

20. **Which stream would you use to copy an image file?**

A: A Byte Stream, like `FileInputStream`, because you must copy exact raw binary data.